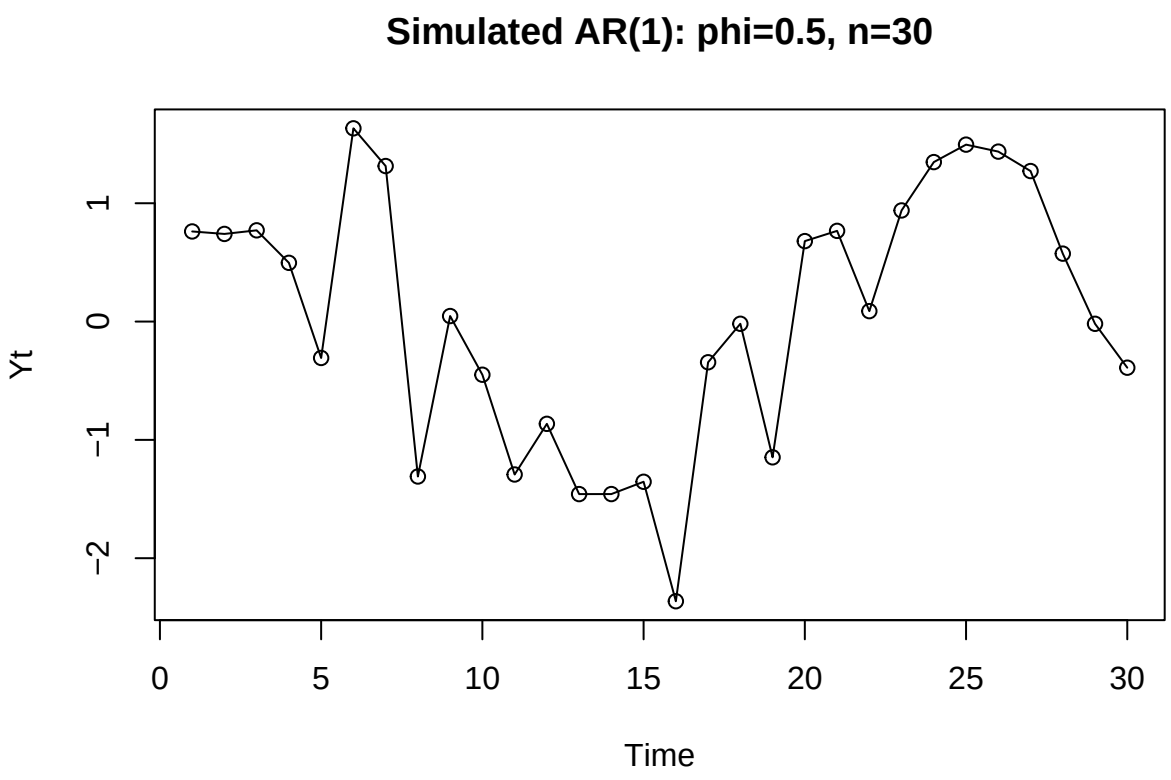
Exercise 9

Mouhamed Cisse

2026-03-15

a)

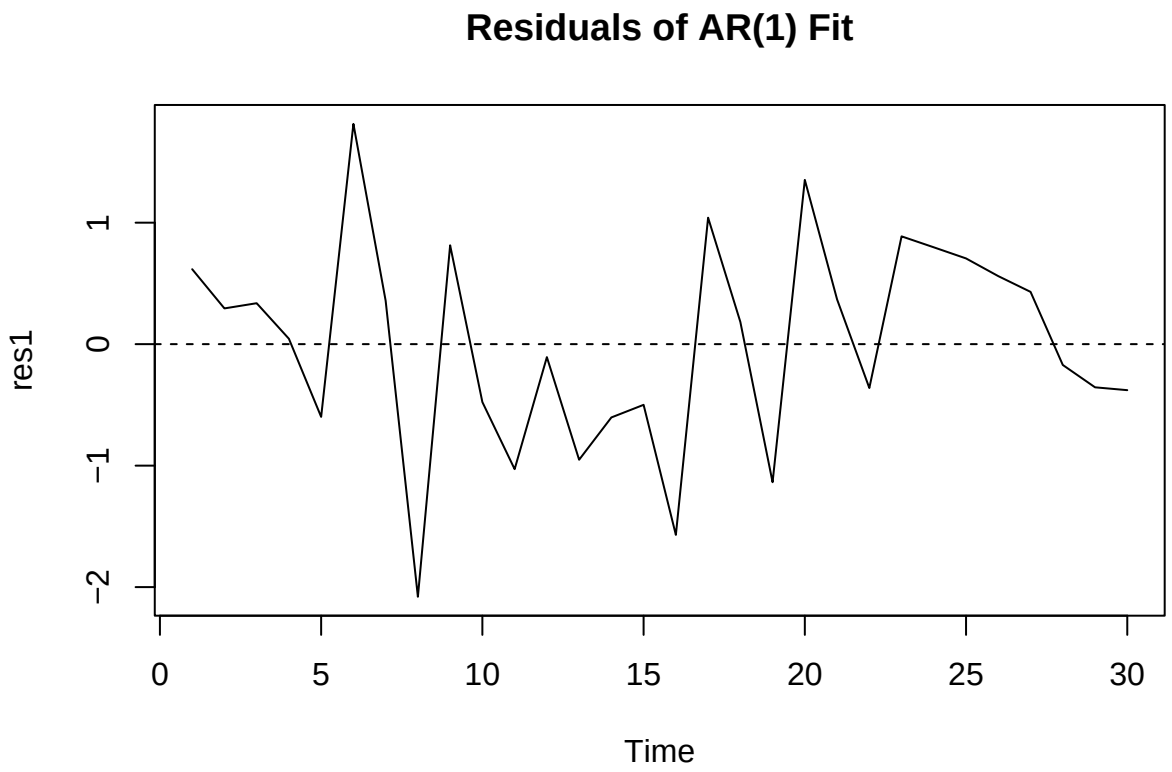
```
set.seed(123)
n1 <- 30
phi <- 0.5
x1 <- arima.sim(n = n1, list(ar = phi))
plot(x1, type = "o", main = "Simulated AR(1): phi=0.5, n=30", ylab = "Yt")
```



b)

```
# Fit model and plot residuals
fit1 <- arima(x1, order = c(1, 0, 0), include.mean = FALSE)
```

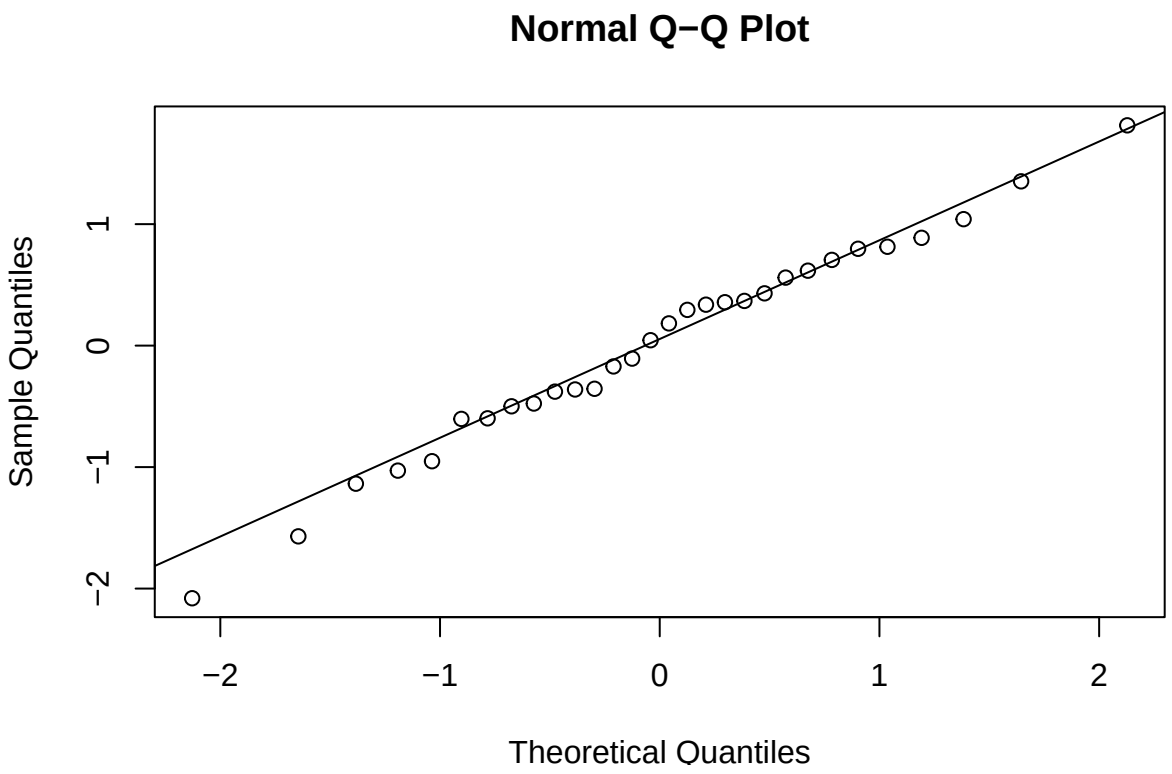
```r
res1 <- residuals(fit1)
plot(res1, main = "Residuals of AR(1) Fit")
abline(h = 0, lty = 2)
```



The residuals seems to behave like random noise, envelopped between -2 and +2.

c)

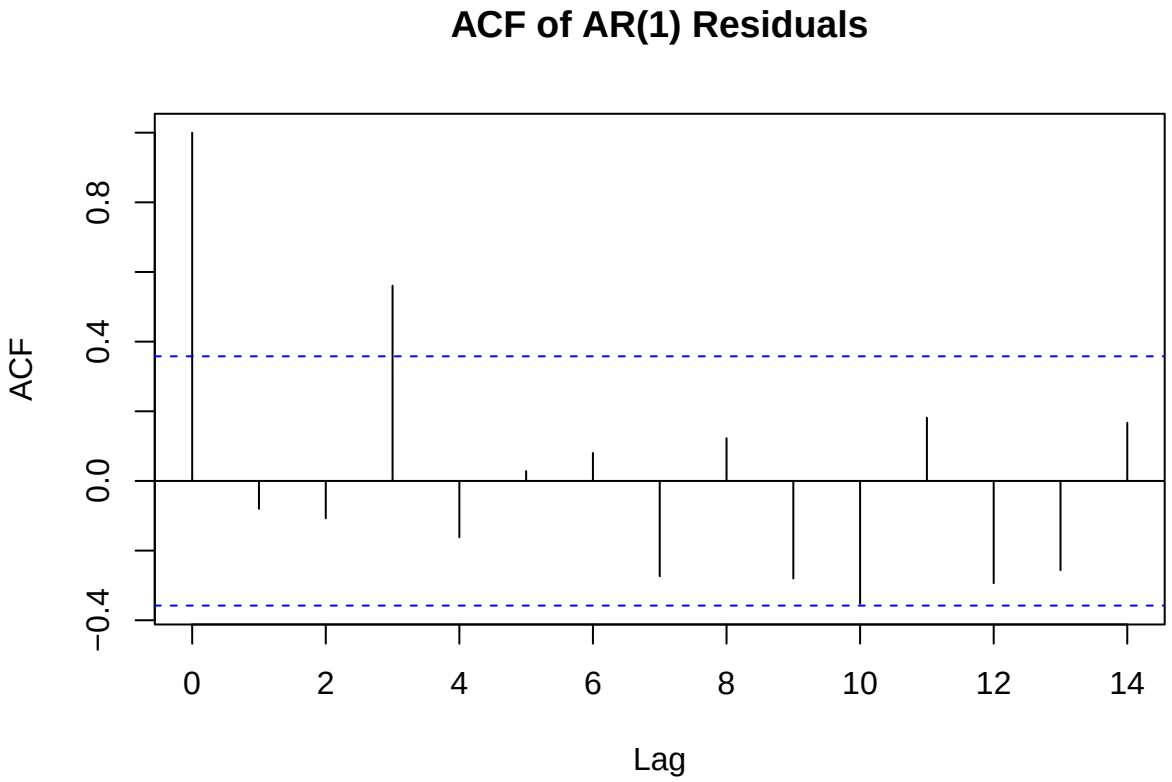
```r
# Normal QQ Plot
qqnorm(res1)
qqline(res1)
```



Most of the points align roughly with the normality line suggestiong a good fit despite some deviation in the tails.

d)

```r
# Sample ACF of residuals
acf(res1, main = "ACF of AR(1) Residuals")
```



The plot does not support the AR(1) specification. In a perfect model, all spikes should stay within the blue dashed lines. A significant spike at Lag 3 suggests that there is still a relationship in the data that the AR(1) didn't capture.

e)

```
# Ljung-Box Statistic (K=8)
Box.test(res1, lag = 8, type = "Ljung-Box", fitdf = 1)
```

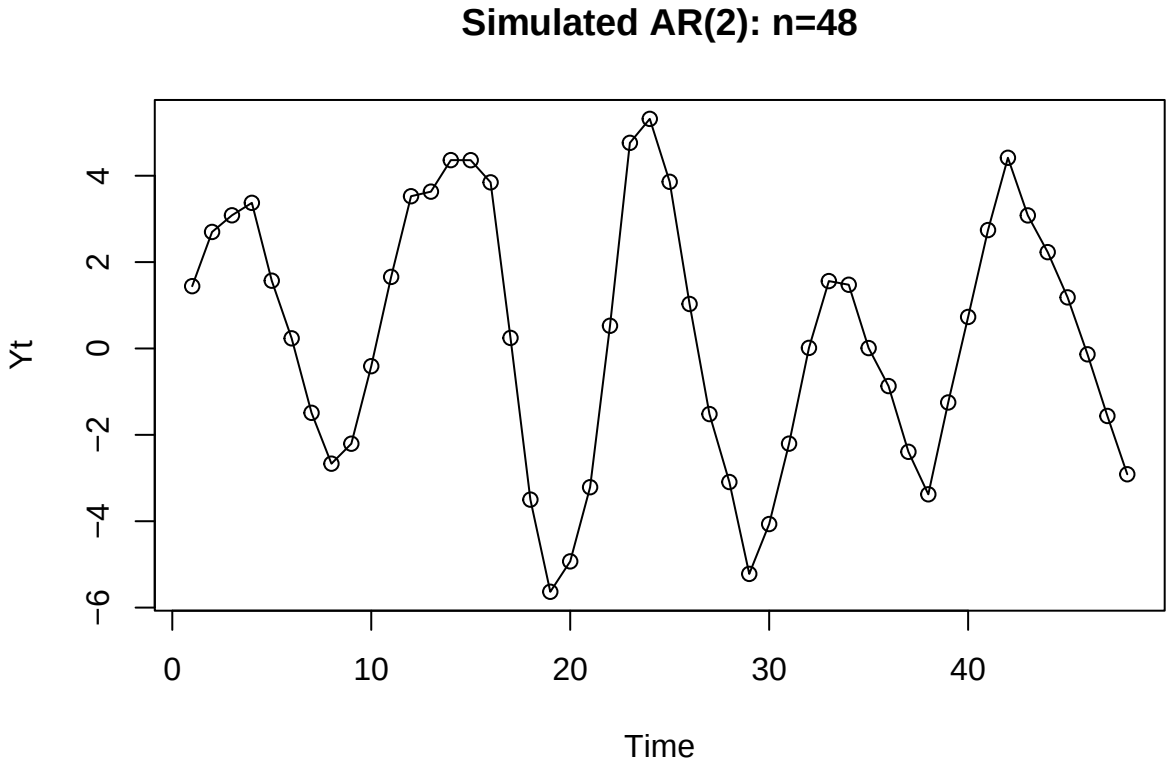
```
##
##  Box-Ljung test
##
## data:  res1
## X-squared = 16.825, df = 7, p-value = 0.01856
```

This does not support the AR(1) specification. Since the p-value (0.018) is less than 0.05, we reject the null hypothesis that the residuals are white noise.

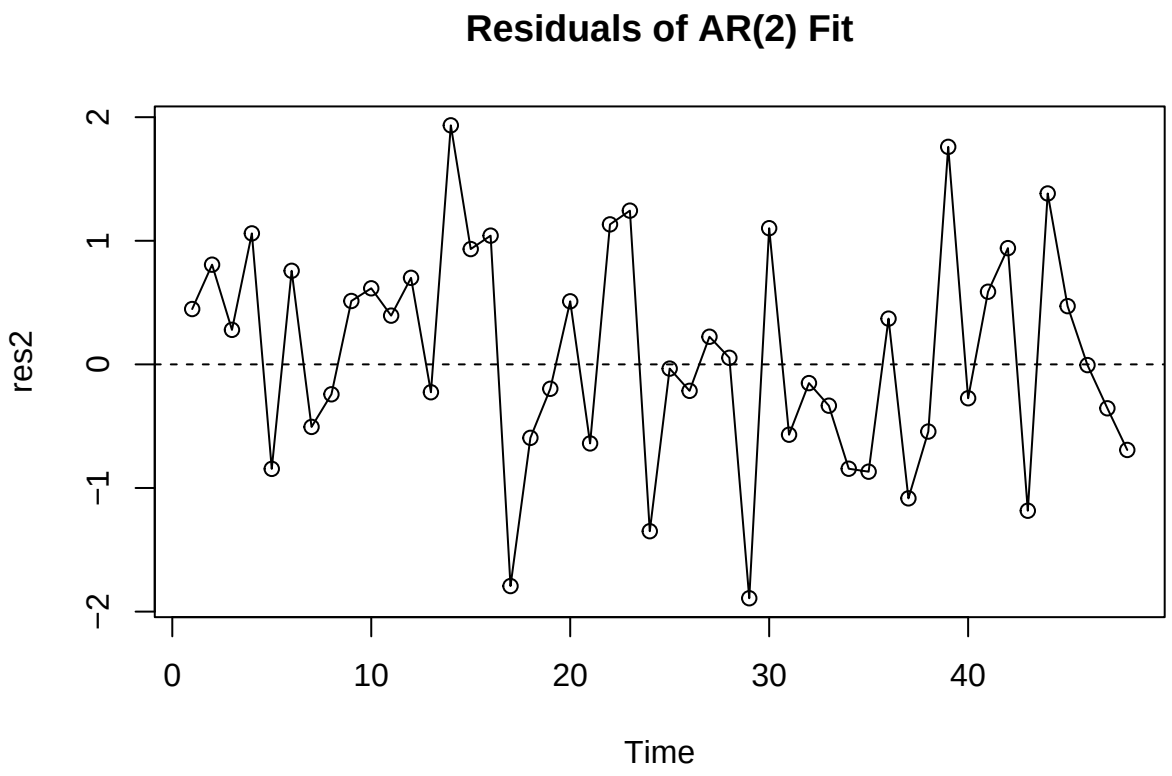
f)

```
set.seed(456)
# (a) Simulate and Plot
n2 <- 48
```

```r
phi_vec <- c(1.5, -0.75)
x2 <- arima.sim(n = n2, list(ar = phi_vec))
plot(x2, type = "o", main = "Simulated AR(2): n=48", ylab = "Yt")
```

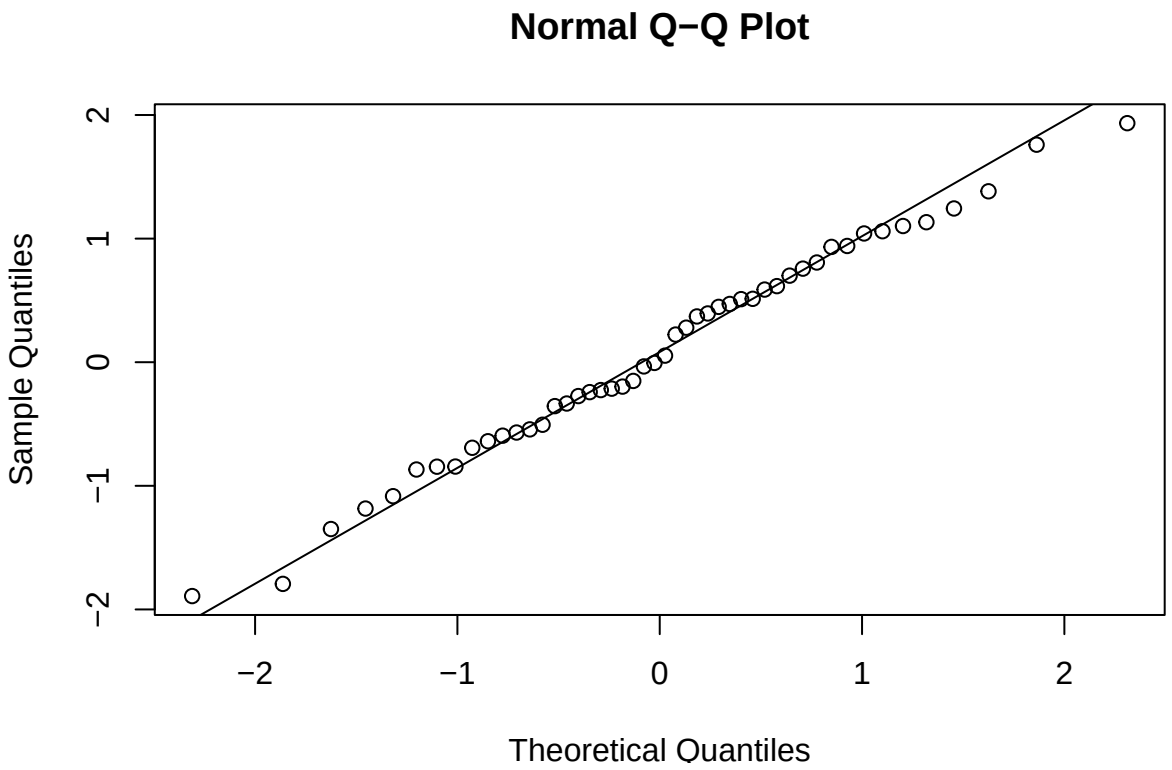



```r
# (b) Fit model and plot residuals
fit2 <- arima(x2, order = c(2, 0, 0), include.mean = FALSE)
res2 <- residuals(fit2)
plot(res2, type = "o", main = "Residuals of AR(2) Fit")
abline(h = 0, lty = 2)
```



The residuals seems to behave like random noise, envelopped between -2 and +2.

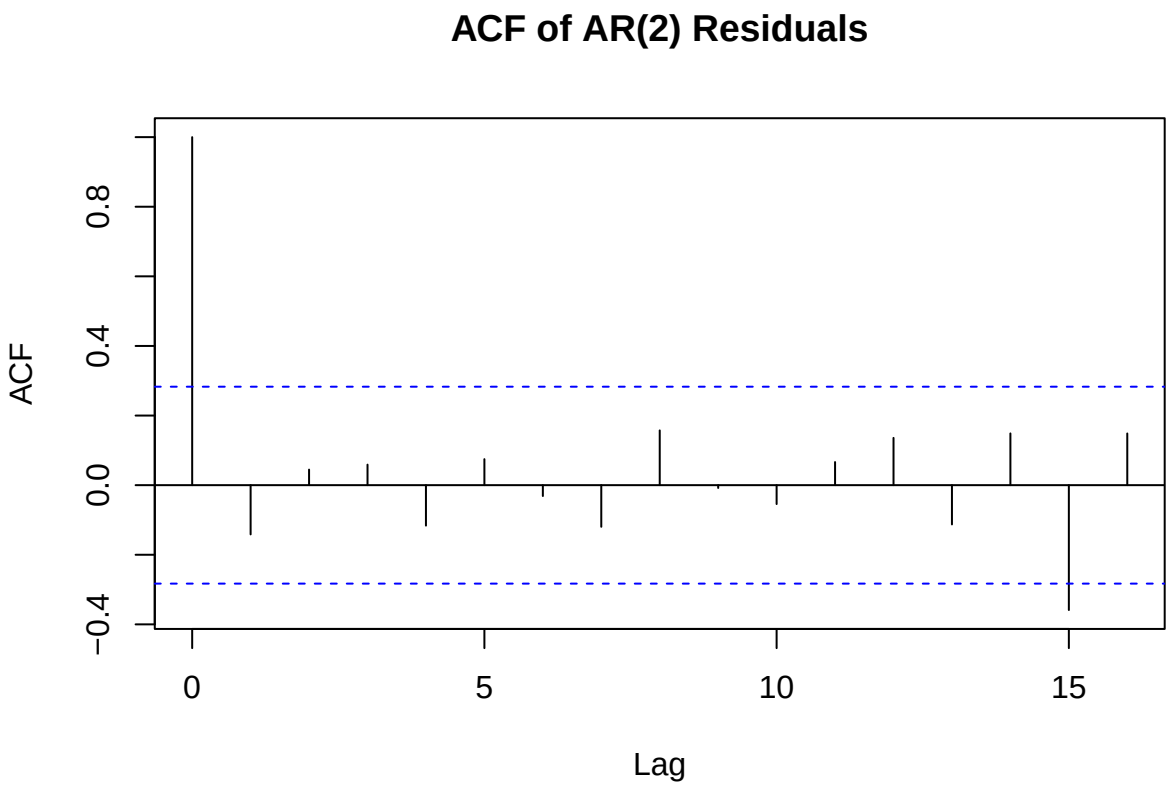
```
# (c) Normal QQ Plot
qqnorm(res2)
qqline(res2)
```



Most

of the points align roughly with the normality line suggestiong a good fit despite some slight deviation in the tails.

```
# (d) Sample ACF of residuals
acf(res2, main = "ACF of AR(2) Residuals")
```



The good fit still hold because only the 15th spike cross the blue dashed line probably due to the small sample size.

```
# (e) Ljung-Box Statistic (K=12)
Box.test(res2, lag = 12, type = "Ljung-Box", fitdf = 2)
```

```
##
##  Box-Ljung test
##
## data:  res2
## X-squared = 6.4506, df = 10, p-value = 0.7761
```

Conclusion:

With a p-value of 0.77, we cannot reject the null hypothesis. This confirms that the residuals are independently distributed (white noise).